

Miscellaneous Antimicrobial Agents

Metronidazole

- ❖ Metronidazole is a nitroimidazole antiprotozoal but
- ❖ it also has potent antibacterial activity against anaerobes
- ❖ Once taken up by anaerobes, it is **nonenzymatically** reduced by reacting with reduced **ferredoxin**.
 - results in products that accumulate in and are toxic to anaerobic cells.
- ❖ The metabolites of metronidazole are taken up into bacterial DNA, forming unstable molecules.

Mechanisms of Metronidazole

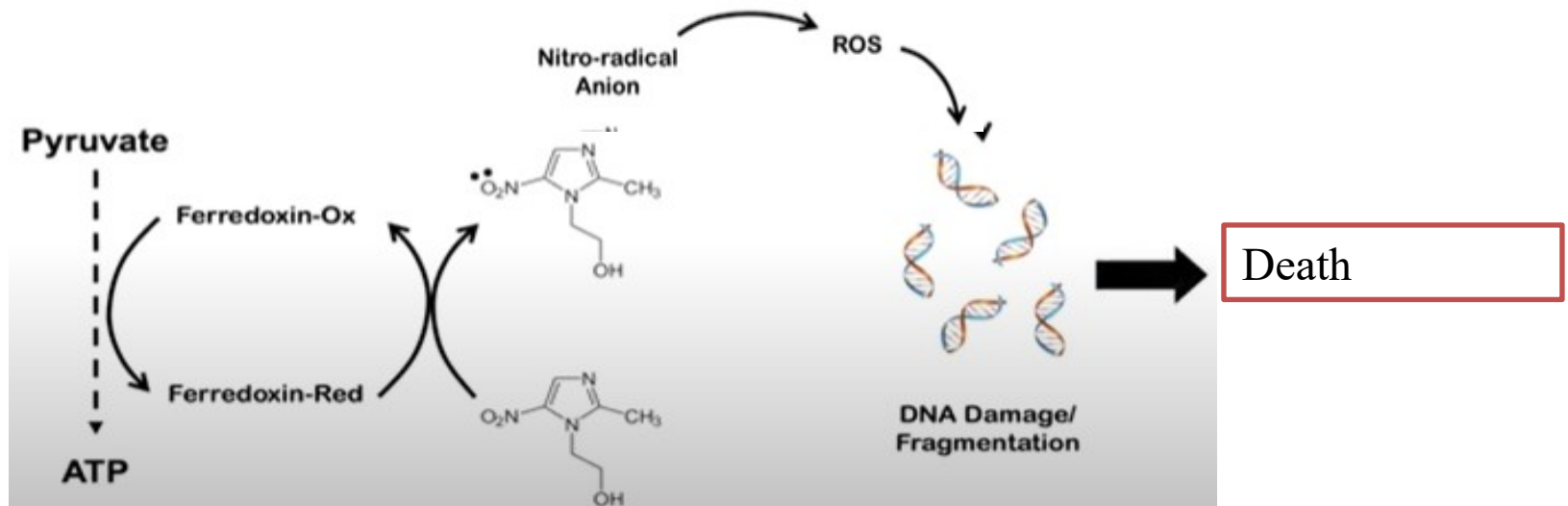


Figure: Mechanisms of Metronidazole

Metronidazole...

- ❖ well absorbed after oral administration
- ❖ It can also be given intravenously.
- ❖ Metronidazole is metabolized in the liver and **may accumulate in hepatic insufficiency.**

Clinical uses

- ❖ for anaerobic or mixed intra-abdominal infections (in combination with other
- ❖ vaginitis (trichomonas infection, bacterial vaginosis),
- ❖ Clostridium difficile infection, and brain abscess.

Dosage... is 500 mg TID orally or IV (30 mg/kg/d).

- ❖ Vaginitis may respond to a single 2-g dose.

Adverse effects

- ❖ nausea, diarrhea, stomatitis, and peripheral neuropathy with prolonged use.
- ❖ Metallic taste
- ❖ Metronidazole has a disulfiram-like effect,
- ❖ Alcohol should be avoided.

Tinidazole

- ❖ Active in anaerobic bacteria
 - approved for treatment of trichomonas infection, giardiasis, amebiasis, and bacterial vaginosis

Fidaxomicin

- ❖ active against Gram-positive aerobes and anaerobes
 - but lacks activity against Gram-negative**
- MOA.....Fidaxomicin inhibits bacterial protein synthesis** by binding to the sigma subunit of **RNA polymerase**.
- approved for the treatment for **C difficile** infection in adults.
- ❖ It is as effective as oral vancomycin with lower rates of **relapsing disease**.
- ❖**Dose:** Fidaxomicin is adm 200 mg tablet BID for 10 days.

Rifaximin

- It is a derivative of rifampin.
- It is active against Gram-positive and Gram-negative aerobes and anaerobes.

MOA.....Rifaximin inhibits protein synthesis by binding to the beta subunit of DNA-dependent RNA polymerase.

- Initially approved for the treatment of travelers' diarrhea,

Currently used for

- ❖ management of hepatic encephalopathy,
- ❖ irritable bowel syndrome with diarrhea, and, occasionally, as an
- ❖ Typical doses of rifaximin range from 200 mg to 550 mg (BID/TID)
- ❖ **not thought to be associated with cytochrome-P450-mediated drug interactions due to its limited absorption.**

Mupirocin

- ❖ It is a natural substance produced by **Pseudomonas fluorescens**.
- ❖ rapidly inactivated after absorption, and systemic levels are undetectable.
 - available as an ointment for topical application.
- ❖ Mupirocin is active against
 - Gram-positive cocci, including MSSA and MRSA

MOA....inhibits staphylococcal isoleucyl tRNA synthetase.

Resistance

Low-level resistance

- due to point mutation in the gene of the target enzyme.
- observed after prolonged use.

High-level resistance

- due to the presence of a second isoleucyl tRNA synthetase gene,
- results in complete loss of activity.
- Strains with high-level resistance have caused hospital-associated outbreaks of staphylococcal infection and colonization.

Clinical uses

- ❖ for topical treatment of minor skin infections, such as **impetigo**
- ❖ Topical application over large open area
 - ✓ leads emergence of mupirocin-resistant strains
 - not recommended.
- ❖ Patients most likely to benefit from decolonization are those undergoing **orthopedic** or **cardiothoracic** procedures.

Polymyxins

- ❖ The polymyxins are a group of basic peptides **active against Gram-negative bacteria** and include
- ❖ polymyxin B and polymyxin E (colistin).

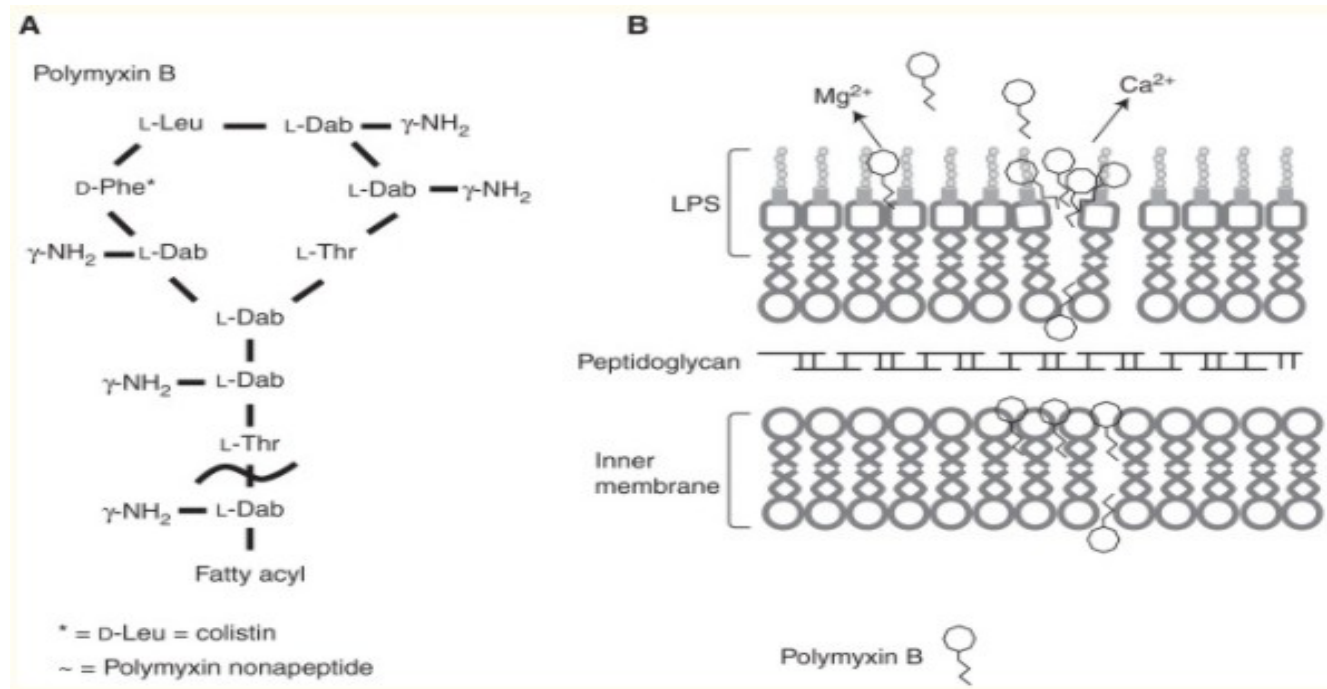


Figure: structure of polymyxin B and likely mode of membrane interaction

Polymyxins...

- ❖ MOA....Polymyxins act as cationic detergents.
- ✓ attach to and disrupt bacterial cell membranes.
- ✓ They also bind and inactivate endotoxin.
- ❖ Available for otic and topical use in combination with a variety of other compounds

Polymyxins...

- ❖ Due to significant toxicity with systemic administration (especially nephrotoxicity),
- ❖ largely restricted to topical use.
- ❖ polymyxin in combination with bacitracin or neomycin (or both) applied to infected **superficial skin lesions**.
- ❖ for salvage therapy of infections caused by
 - strains of *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacteriaceae*

Nitrofurantoin

- ❖ At therapeutic doses, it is **bactericidal** for many **G(-)** and **G(+)** but **P aeruginosa** and many strains of **Proteus** are inherently resistant.
- ❖ **MOA,,,,,**has a complex mechanism of action
Antibacterial activity correlate with
 - ❖ rapid intracellular conversion of nitrofurantoin to **highly reactive intermediates** by bacterial reductases.
 - ❖ intermediates react nonspecifically with many ribosomal proteins
 - ❖ disrupt metabolic processes and the synthesis of proteins, RNA, and DNA.

Nitrofurantoin...

- ❖ There is no cross-resistance between **nitrofurantoin** and other antimicrobial agents,
- ❖ resistance emerges slowly.
 - nitrofurantoin has become an important alternative oral agent for treatment of **uncomplicated urinary tract infection**.

Nitrofurantoin...

- ❖ In renal failure, **urine levels** are insufficient for antibacterial action, but
- ❖ high blood levels may cause toxicity.
 - C/I in a significant renal insufficiency.
- ❖ If CrCl < 60 mL/min....better to avoid
- ❖ short-term nitrofurantoin treatment is with creatinine clearance > 30 mL/min is acceptable

Nitrofurantoin...

- ❖ for UTI100 mg QID
- ❖ A single daily dose of nitrofurantoin, 100 mg, can prevent recurrent urinary tract infections in some women.

Forms of Nitrofurantoin

- ❖ Nitrofurantoin is found in **microcrystalline form** (nitrofurantoin monohydrate); or
- ❖ a macrocrystalline formulation
- ❖ **Currently** most formulations are either purely macrocrystalline or mixed (25% macrocrystalline/75% microcrystalline)

Nitrofurantoin...

- ❖ The drug should not be used to treat upper **urinary tract infection**.
- ❖ pH below 5.5, which greatly enhances drug activity.
- ❖ A single daily dose of nitrofurantoin, **100 mg**, can prevent recurrent urinary tract infections in some women.

Nitrofurantoin...

- Anorexia, nausea, and vomiting are the principal side effects of nitrofurantoin.
- Neuropathies and pulmonary toxicities
 - with prolonged use or
 - in patients with renal impairment.
- ❖ Hemolytic anemia can occur in patients with glucose-6-phosphate dehydrogenase deficiency.
- ❖ Nitrofurantoin antagonizes the action of **nalidixic acid**.